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COMM022 – Machine Learning Algorithms

Comparative Analysis of Classical Machine Learning Algorithms for Terrain Image Classification

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MSC ROBOTICS

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Abstract

Terrain classification plays an important role in robotics and autonomous navigation, as different ground surfaces can affect movement, stability, and decision-making. In this project, a machine learning based approach is used to classify terrain images into five categories: grassy, sandy, rocky, marshy and other. A labelled image dataset was collected and organized, and a complete processing pipeline was developed, starting from data preparation to model evaluation. Several classical machine learning models were implemented, including Support Vector Machine (SVM), Random Forest, and K-Nearest Neighbors (KNN). Image features were first extracted and then used to train and test each model under the same conditions. The performance of the models was evaluated using accuracy, macro F1-score, and confusion matrices in order to understand both overall performance and class-level behavior. The results show that the Random Forest model achieved the best performance among the three models, while SVM also produced stable and reliable results. KNN performed relatively poorly, mainly due to the high dimensional feature space. This study demonstrates that traditional machine learning methods can effectively classify terrain images and can be suitable for practical applications where computational resources are limited.

1. Introduction

Terrain classification is an important task in computer vision and robotics, where the aim is to identify different types of ground surfaces such as grass, sand, rocky areas, and marshy regions using visual information. This task is particularly relevant for autonomous robots and vehicles that operate in outdoor environments, as understanding the terrain can support safer navigation, improved path planning, and better decision-making. In real-world conditions, terrain appearance can vary significantly due to changes in lighting, texture, and surface conditions, which makes reliable classification a challenging problem.

Traditional rule-based methods struggle to handle these variations, as it is difficult to manually define features that generalize well across different environments. Machine learning offers a more flexible approach by learning patterns directly from data rather than relying on predefined rules. In image-based classification tasks, numerical features extracted from images are used to represent visual characteristics that help distinguish between terrain types. However, the effectiveness of such systems depends heavily on the choice of learning algorithm, as different models may respond differently to the same feature set.

In this project, a real-world terrain image dataset is used to evaluate the performance of three classical machine learning algorithms: Support Vector Machine, Random Forest, and k-Nearest Neighbour. These algorithms were selected because they represent different learning strategies and are commonly used in practical applications. By training and testing each model under the same conditions, this study aims to provide a clear comparison of their suitability for terrain classification and to understand the strengths and limitations of each approach.

2. Aims and Objectives

The aim of this project is to design and evaluate machine learning models for classifying terrain images into different terrain categories using a real-world dataset. The focus is on understanding how well classical machine learning algorithms perform when applied to the same classification task under consistent conditions.

To achieve this aim, the project involves organizing and preprocessing a terrain image dataset, followed by the extraction of

meaningful numerical features that capture important visual characteristics of the surfaces. Three machine learning classifiers, namely Support Vector Machine, Random Forest, and k-Nearest Neighbour, are then trained using the same feature representation. The performance of these models is evaluated using accuracy, macro F1-score, and confusion matrices. Finally, the results are compared to identify the most effective model and to analyze the limitations and strengths of each approach in a practical terrain classification scenario.

3. Literature Review

Terrain classification using visual information has become an active research area, particularly for robotic systems that operate in outdoor environments. (Wang, et al., 2022) studied terrain classification for mobile robot navigation using RGB camera images. Their work showed that terrain surfaces such as grass, sand, and rocky areas often appear similar in colour and texture, which makes classification difficult. The authors also reported that classification accuracy depends strongly on the choice of learning algorithm, especially when terrain classes overlap visually. These challenges are relevant to the present project, as similar confusion between terrain categories was observed during the evaluation of the machine learning models.

(Chen, et al., 2021) focused on terrain classification for unmanned ground vehicles operating in real outdoor conditions. Their study highlighted the impact of environmental factors such as lighting changes, uneven surfaces, and noise on classification performance. Although their approach combined visual data with additional sensor information, the authors emphasized that visual terrain classification remains a core component of reliable navigation systems. They also noted that some terrain classes are more difficult to distinguish than others due to visual similarity. This supports the motivation of the current project, which evaluates different machine learning algorithms to understand how they respond to the same terrain image data.

(Hanson, et al., 2022) presented a terrain classification framework designed for unstructured and multi-class outdoor environments using visual data collected in realistic settings. The authors argued that real-world datasets better represent the complexity of natural terrain surfaces compared to controlled laboratory data. This view is closely aligned with the dataset used

in this project, which consists of terrain images captured under natural conditions. Their work highlights the importance of evaluating terrain classification methods in realistic environments where surface variation and noise are unavoidable.

While many recent studies focus on deep learning approaches or sensor fusion techniques to improve terrain classification performance, such methods often require large datasets and higher computational resources. In contrast, this project focuses on classical machine learning algorithms trained on extracted image features. By evaluating Support Vector Machine, Random Forest, and k-Nearest Neighbour classifiers using the same dataset and feature representation, this study aims to provide a clear comparison of traditional models under identical conditions. This approach is particularly suitable for scenarios where computational efficiency and simplicity are important, such as robotics applications with limited resources.

4. Dataset Description and Preprocessing

The dataset used in this project contains 1,989 RGB images divided into five terrain categories: grassy, sandy, rocky, marshy, and other terrain types. Each category includes a similar number of images, which helps reduce the risk of bias during model training and allows a fair comparison between classifiers. The images were collected from real outdoor environments and show natural variations in colour, texture, and surface appearance. This dataset was selected because it represents realistic conditions faced by robotic systems operating outdoors, where terrain recognition is important for navigation and safety. In addition, the visual similarity between some terrain classes makes the classification task more challenging and suitable for evaluating the performance of different machine learning models.

Before applying the machine learning algorithms, the dataset was carefully prepared through several preprocessing steps. First, the dataset structure was checked to ensure that all images were correctly organized into their respective class folders and that no files were missing or corrupted. Since the original images were high in resolution, they were resized to a fixed size to reduce computational cost while still retaining important visual information. This also ensured consistency across all samples. The preprocessing stage was carried out to make the images suitable for feature

extraction, as classical machine learning models require numerical input rather than raw image data. These steps helped reduce unnecessary variation in the dataset and produced a clean and structured input for feature extraction and model training, supporting more reliable evaluation of the classifiers used in this project.

5. Feature Extraction

Feature extraction was a necessary step in this project because classical machine learning models cannot directly use raw image pixels as input. Instead, the visual information in each terrain image needs to be converted into numerical values that represent important surface characteristics. The main focus during this stage was to capture information related to texture, edges, and general surface patterns, as these properties vary noticeably between terrain types such as grass, sand, and rocky surfaces. Using a feature-based representation made it possible to describe each image in a consistent way and provided suitable input for the machine learning classifiers.

Each preprocessed image was therefore processed to extract features that describe texture and gradient-related information. These features were chosen because they are well suited to representing surface variations commonly found in outdoor terrain images. Once extracted, the features were stored as fixed-length numerical vectors and combined to form the final dataset used for training and testing the models. Since some classifiers are sensitive to differences in feature scale, feature scaling was applied where required to ensure fair comparison between models. Overall, this feature extraction step helped the models focus on meaningful visual patterns rather than raw pixel values, which contributed to more stable training and clearer interpretation of the classification results.

6. Methodology

This project was implemented locally using Python, with all coding and testing carried out in Visual Studio Code. A number of standard Python libraries were used during development, including NumPy for handling numerical data, OpenCV for reading and processing images, scikit-learn for training and evaluating machine learning models, and Matplotlib for visualising results. After completing the preprocessing and feature extraction stages, the extracted features were combined with their corresponding class labels to form a single dataset. To evaluate the models in a consistent way, the dataset

was split into training and testing sets using an 80:20 ratio. This allowed the models to be trained on most of the data while still being tested on unseen samples, which helped assess how well they generalized to new terrain images.

Three classical machine learning classifiers were implemented in this project: Support Vector Machine, Random Forest, and k-Nearest Neighbour. These models were chosen because they represent different learning approaches and are commonly used in image classification tasks. The same set of extracted features was used for all three models to ensure a fair comparison. Feature scaling was applied before training the SVM and KNN models, as these algorithms are sensitive to differences in feature values, while Random Forest was trained without scaling since it is not affected by feature magnitude. After training, the models were evaluated using accuracy, macro F1-score, and confusion matrices. These evaluation measures were selected to provide both an overall view of performance and insight into how well individual terrain classes were classified. The trained models were saved after evaluation so that the results could be reproduced if required. This methodology made it possible to compare the three classifiers clearly and consistently for the terrain classification task.

7. Results and Discussion

This section discusses the results obtained from the three machine learning models used in this project: Support Vector Machine (SVM), Random Forest, and k-Nearest Neighbour (KNN). The performance of each model was evaluated using accuracy, macro F1-score, and confusion matrices. These evaluation methods were chosen to understand not only the overall performance but also how well each model handled individual terrain classes.

7.1 Overall Performance of the Models

The overall performance comparison of the models is shown in Figures 7.1 and 7.2, where accuracy and macro F1-score are compared. From the results, Random Forest achieved the best performance with an accuracy of 87.69% and a macro F1-score of 0.8748. The SVM model also performed reasonably well, achieving an accuracy of 81.66% and a macro F1-score of 0.8151. In contrast, the KNN model performed significantly worse, with an accuracy of 48.74% and a macro F1-score of 0.4369.

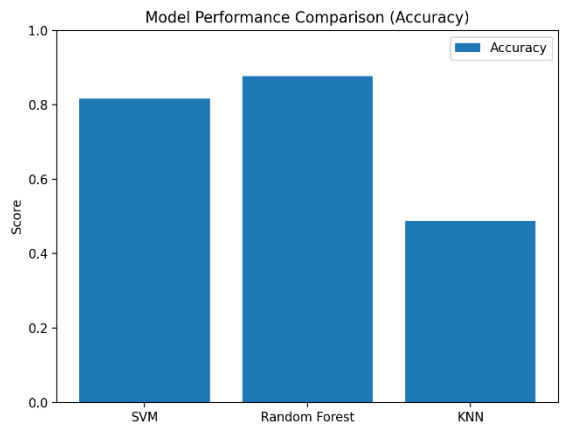

Figure 7.1: Accuracy comparison of SVM, Random Forest, and KNN models on the terrain classification dataset.

The strong performance of the Random Forest model can be linked to its ensemble-based nature. Since it uses multiple decision trees, it can learn complex patterns in the feature data and reduce the effect of noise. This was particularly useful for this dataset,

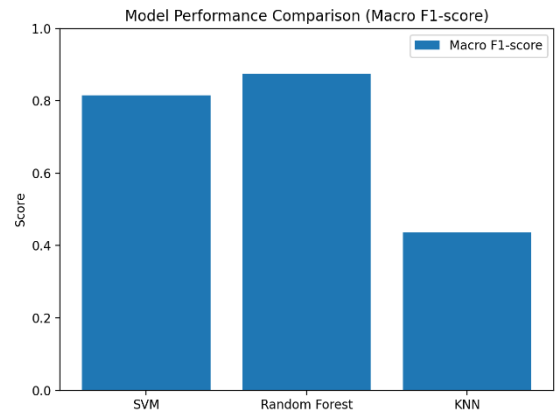

Figure 7.2: Macro F1-score comparison of SVM, Random Forest, and KNN models on the terrain classification dataset.

where terrain images contain natural variations in texture and appearance. The SVM model also performed well because it can handle high-dimensional feature spaces effectively. However, its performance was slightly lower than Random Forest, possibly due to overlapping features between some terrain classes. The poor performance of KNN suggests that distance-based methods are not well suited for this task, especially when working with high-dimensional image features.

7.2 Confusion Matrix Analysis

To better understand how each model behaved at the class level, confusion matrices were analyzed, as shown in Figures 7.3, 7.4, and 7.5. The Random Forest confusion matrix shows that most predictions fall along the diagonal, which means that the majority

of terrain images were classified correctly. Terrain types such as Sandy and Other were recognized very accurately, showing that Random Forest was able to capture clear patterns from the extracted features.

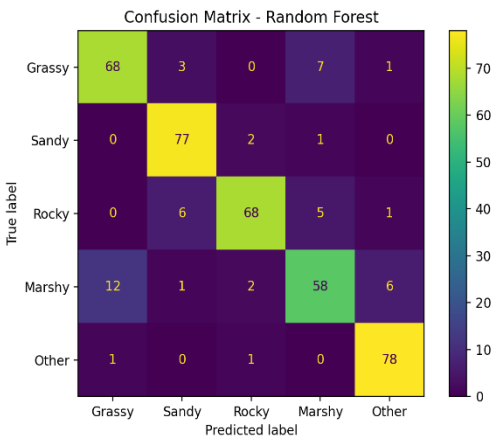

Figure 7.3: Confusion matrix for the Random Forest model showing classification performance across terrain classes.

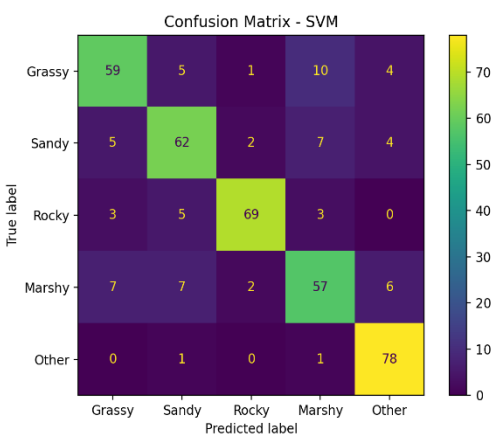

Figure 7.4: Confusion matrix for the Support Vector Machine (SVM) model showing terrain classification results.

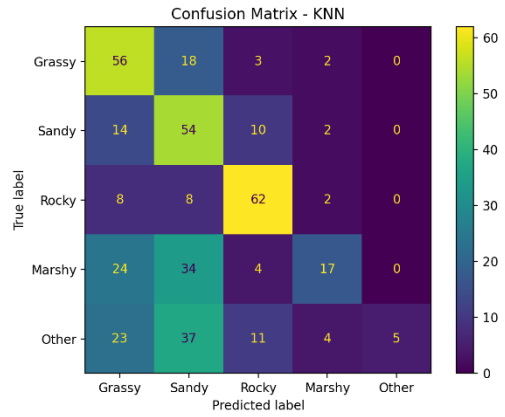

Figure 7.5: Confusion matrix for the K-Nearest Neighbours (KNN) model showing terrain classification results.

The SVM confusion matrix also shows good performance for classes like Rocky and Other. However, there was noticeable confusion between Grassy and Marshy terrains. This can be explained by the visual similarity between these two terrain types. Both often contain green textures and uneven surfaces, which makes them difficult to distinguish using traditional feature-based methods. This suggests that while SVM is effective, it may struggle when class boundaries are not clearly defined in the feature space.

The KNN confusion matrix shows high levels of misclassification across most terrain classes. Many samples were incorrectly predicted as Grassy or Sandy, regardless of their true label. This behavior indicates that KNN was unable to form clear decision

boundaries in this dataset. Since KNN relies on distance calculations, the high dimensionality of the feature vectors likely reduced its effectiveness, leading to poor generalization.

7.3 Comparative Discussion

When comparing all three models, Random Forest clearly performed the best in terms of accuracy, class balance, and overall reliability. Its ability to handle non-linear relationships and noisy features made it well suited for terrain classification using extracted image features. The SVM model served as a strong baseline and produced stable results, but it showed limitations when dealing with visually similar terrain types. KNN performed the weakest, highlighting the limitations of simple distance-based classifiers for complex image-based classification problems.

The combined performance comparison shown in Figure 7.6 further supports these findings. Both accuracy and macro F1-score follow the same trend, with Random Forest achieving the highest values, followed by SVM and then KNN. These results show that model choice has a significant impact on performance, even when all models are trained on the same dataset using the same features. Overall, the results demonstrate that ensemble learning methods are more suitable for this type of terrain classification task, especially in realistic outdoor environments where visual patterns often overlap.

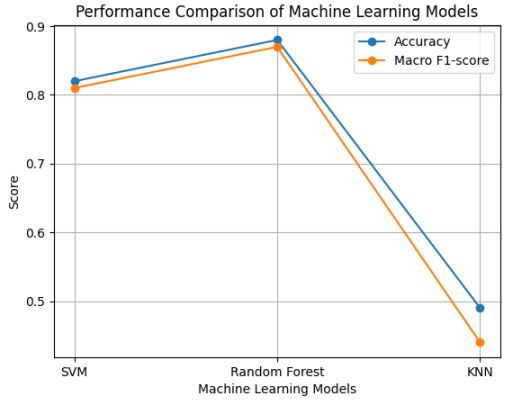

Figure 7.6: Line graph comparing accuracy and macro F1-score of SVM, Random Forest, and KNN models.

8. Conclusion

This project focused on applying classical machine learning methods to the problem of terrain image classification using a real-world dataset. The aim was to compare how different classifiers perform when trained on the same features and tested under the same conditions. Three models were implemented in this work: Support Vector Machine, Random Forest, and k-Nearest Neighbour. Based on the results obtained, the Random Forest model showed the best overall performance, achieving the highest accuracy and macro F1-score across most terrain classes.

The Support Vector Machine also performed well, but some misclassification was observed between terrain types that share similar visual characteristics, such as grassy and marshy surfaces. This shows that visual similarity between classes remains a challenge for image-based terrain classification. The k-Nearest Neighbour model produced lower performance compared to the other two methods, especially when dealing with overlapping features. Overall, the project met its objectives by successfully building a complete machine learning pipeline, evaluating different classifiers, and analyzing their strengths and limitations. The results demonstrate that classical machine learning approaches can still be effective for terrain classification when appropriate features and evaluation methods are used.

9. Future Work

There are several ways this project could be improved or extended in the future. One possible direction is to use deep learning models, such as convolutional neural networks, which are capable of learning features directly from raw images. This could reduce the need for manual feature extraction and may improve classification performance for visually complex terrain types. Increasing the size of the dataset or applying data augmentation techniques could also help the models generalize better to unseen data.

Another possible improvement would be to test the trained models in a simulated or real robotic environment. This would allow the evaluation of model performance under practical conditions, where factors such as real-time processing and environmental changes are important. Further experimentation with parameter tuning or combining different approaches could also be explored. These extensions would help move

the work closer to real-world applications in robotics and autonomous navigation.

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